



IC CHIP DESIGN AND VERIFICATION

“Design of Washer and Dryer Controller”

A PROJECT REPORT

Submitted by

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*in partial fulfilment for the award of the degree
of*

**BACHELOR OF ENGINEERING
IN
ELECTRONICS AND COMMUNICATION ENGINEERING**

MAHENDRA COLLEGE OF ENGINEERING, SALEM

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DEC – (10.12.2025-23.12.2025)

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the **Maven Silicon team** for providing an excellent online training program on **IC Chip Design and Verification**, through which I gained strong conceptual and practical knowledge in digital design and verification concepts. The well-structured sessions and learning materials, including the project-based approach, greatly helped me in understanding and implementing the concepts presented in this project.

I express my heartfelt thanks to **Ms. Sandhya** for her valuable guidance, continuous support, and encouragement throughout the project session. The entire project work was carried out through **online sessions**, and her clear explanations and effective teaching made the learning process smooth and productive.

I extend my sincere thanks to **Dr. Sugandhi, Ph.D., Head of the Department**, for her constant encouragement and support in completing this project successfully.

I would like to convey my deep sense of gratitude to **Dr. N. Mohanasundararaju, Principal**, for providing the necessary facilities and support to carry out this project.

I also express my sincere thanks to my **Class Advisor, Mrs. S. Meera**, for her motivation, guidance, and continuous encouragement throughout the course of this project.

Finally, I would like to thank my parents, friends, and all those who directly or indirectly supported me during the successful completion of this project.

ABSTRACT

The rapid advancement of digital systems has significantly increased the demand for reliable, efficient, and automated control mechanisms in embedded and electronic applications. One such widely used application is the washing machine, which requires precise control of multiple sequential operations such as washing, rinsing, and drying. This project focuses on the **design and verification of a Washing Machine Controller using Finite State Machine (FSM) concepts**, implemented through **Register Transfer Level (RTL) design** techniques.

The primary objective of this project is to model the functioning of an automatic washing machine controller that operates based on various input conditions and sensor signals. The controller initiates operation upon receiving a coin input and proceeds through a sequence of well-defined states including **Idle, Counter, Wash, Rinse, and Dry**. Each state is responsible for controlling specific operations such as water filling, washing duration, rinsing cycles, and drying, ensuring an efficient and systematic flow of the washing process.

The design incorporates multiple input signals such as **coin input, soap sensor, and wetness sensor**, which dynamically influence the state transitions of the controller. An internal counter is utilized to monitor the washing duration, ensuring that the wash cycle is completed for a fixed time period of thirty minutes. The rinsing and drying operations are conditionally controlled based on sensor feedback, thereby improving automation and reliability.

The entire system is designed using **Verilog Hardware Description Language (HDL)** following the FSM-based design methodology. A detailed **state transition diagram, state transition table**, and **state encoding scheme** are developed to accurately represent the system behavior. The RTL design is verified through a comprehensive **testbench**, and the functionality of the controller is validated using **simulation waveforms**. Furthermore, the design is synthesized to obtain the **synthesis schematic**, confirming the logical correctness and hardware feasibility of the controller.

This project was carried out as part of an **online training session conducted by the Maven Silicon team**, providing hands-on exposure to industrial-level digital design and verification practices. The successful implementation and verification of the washing machine controller demonstrate the effectiveness of FSM-based control systems and highlight the practical application of digital design concepts in real-world automation systems.

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INTRODUCTION

In recent years, the rapid growth of digital electronics and automation has led to the widespread adoption of intelligent control systems in everyday appliances. Automatic washing machines are one of the most commonly used household devices that rely heavily on digital control mechanisms to perform complex sequential operations efficiently. The design of such control systems requires precise coordination between hardware components, sensors, and control logic to ensure reliable and automated functioning.

A washing machine controller is a digital system responsible for managing various operations such as washing, rinsing, and drying based on predefined conditions and user inputs. Traditional mechanical control systems lack flexibility and accuracy, whereas digital controllers provide better reliability, scalability, and ease of modification. Finite State Machines (FSMs) play a crucial role in designing such controllers, as they allow the system to move through a series of well-defined states based on input conditions and sensor feedback.

This project focuses on the **design and verification of a Washing Machine Controller using Finite State Machine concepts**, implemented at the **Register Transfer Level (RTL)** using **Verilog Hardware Description Language (HDL)**. The controller operates by receiving inputs from various sensors such as coin input, soap sensor, and wetness sensor, and generates appropriate control signals to manage the washing process automatically.

The washing machine controller designed in this project consists of five major states: **Idle, Counter, Wash, Rinse, and Dry**. The system initially remains in the Idle state, waiting for a coin input to start the washing cycle. Upon receiving the coin input, the controller transitions through the Counter state, which initiates an internal timer. The Wash state performs the washing operation for a fixed duration of thirty minutes using an internal counter. After completion, the system moves to the Rinse state until the soap sensor turns off. Finally, the Dry state continues until the wetness sensor indicates dry clothes, after which the system returns to the Idle state.

The design methodology adopted in this project follows a structured digital design flow, including requirement analysis, state diagram creation, state table generation, state encoding, RTL coding, testbench development, simulation, and synthesis. This project was carried out as part of an **online IC Chip Design and Verification training program conducted by the Maven Silicon team**, providing practical exposure to FSM-based digital system design and verification.

OBJECTIVE OF THE PROJECT

The main objective of this project is to **design and verify a Washing Machine Controller using Finite State Machine (FSM) concepts** at the **Register Transfer Level (RTL)**. The project aims to model an automatic washing process that performs washing, rinsing, and drying operations in a sequential and controlled manner based on sensor inputs and timing constraints.

The specific objectives of the project are as follows:

- To understand and apply **Finite State Machine (FSM)** concepts in digital system design.
- To design a washing machine controller that operates based on **coin input, soap sensor, and wetness sensor**.
- To implement the controller using **Verilog Hardware Description Language (HDL)**.
- To incorporate an internal counter to control the washing duration effectively.
- To develop a **testbench** for functional verification of the design.
- To verify the correctness of the design using **simulation waveforms**.
- To perform **synthesis** and analyze the synthesis schematic for hardware feasibility.
- To gain practical exposure to **IC chip design and verification methodologies** through an online training program conducted by the Maven Silicon team.

PROBLEM STATEMENT

Design a digital circuit that acts as a controller to a washer and dryer machine with following specifications.

- The machine should perform the following operation.
- Receive a coin to the wash cycle.
- Wash the cloths for 30 minutes.
- Rinse the cloths. If the soap in the cloths is still present, it should continue to rinse.
- Dry the clothes.
- The machine takes in input from coin, soap, and wetness sensor to perform the operation.
- The wash operation is carried out for 30 minutes using the internal counter.

SYSTEM SPECIFICATIONS

The machine should perform the following operations:

- Receive a coin to start the wash cycle.
- Wash the clothes for 30 minutes.
- Rinse the clothes.
- If the soap in the clothes is still present, it should continue to rinse. Dry the clothes.
- This machine takes in inputs from coin , soap and wetness sensors to perform the above operations.
- The wash operation is carried out for 30 minutes using the internal counter.

Inputs	Outputs
Coin put in (coin=1)	(Coin=1) will initiate a counter and Timer begin indicator (T=1), Fill water (FW=1)
Soap is present (Soap=1)	Drain and Refill (DR=1)
Wetness indicator (Wet=1)	Drain completely (D=1)

State	State Description
Initial State	- The machine is in an initial state that is waiting for coin to be inserted. - All the outputs are zero. - When the coin is inserted (coin=1), transition from “initial state” to “counter state” happens
Counter	- This state will output the timer begin signal (T=1) - This will initiate the counter. - Once the counter is initiated, unconditionally the transition happens from “counter state” to “wash state”.
Wash	- This will output a signal to fill the water for washing clothes (FW=1) - Washing of clothes will continue for 30 mins (30 clock pulses). - Once the count is more than 30, transition from “wash state” to “Rinse state” happens.
Rinse State	- This will output a signal to refill with new water for rinsing (R = 1). - As long as the soap sensor is on (Soap = 1), the cycle continues. - When the soap sensor turns off (i.e. Soap = 0), this state will end and thus transition from “Rinse State” to “Dry State”.
Dry State	- This will output a signal to drain the water for good and begin drying (D = 1). - As long as the wet clothes sensor is on (Wet = 1), the cycle continues. - When the wet clothes sensor is off (Wet = 0), the cycle will stop and goes back to initial state.

DIGITAL DESIGN PROCESS

The digital design process adopted for the washing machine controller follows a systematic and structured approach to ensure correct functionality and reliable operation. The design begins with understanding the system requirements and identifying the required inputs, outputs, and operating conditions of the washing machine.

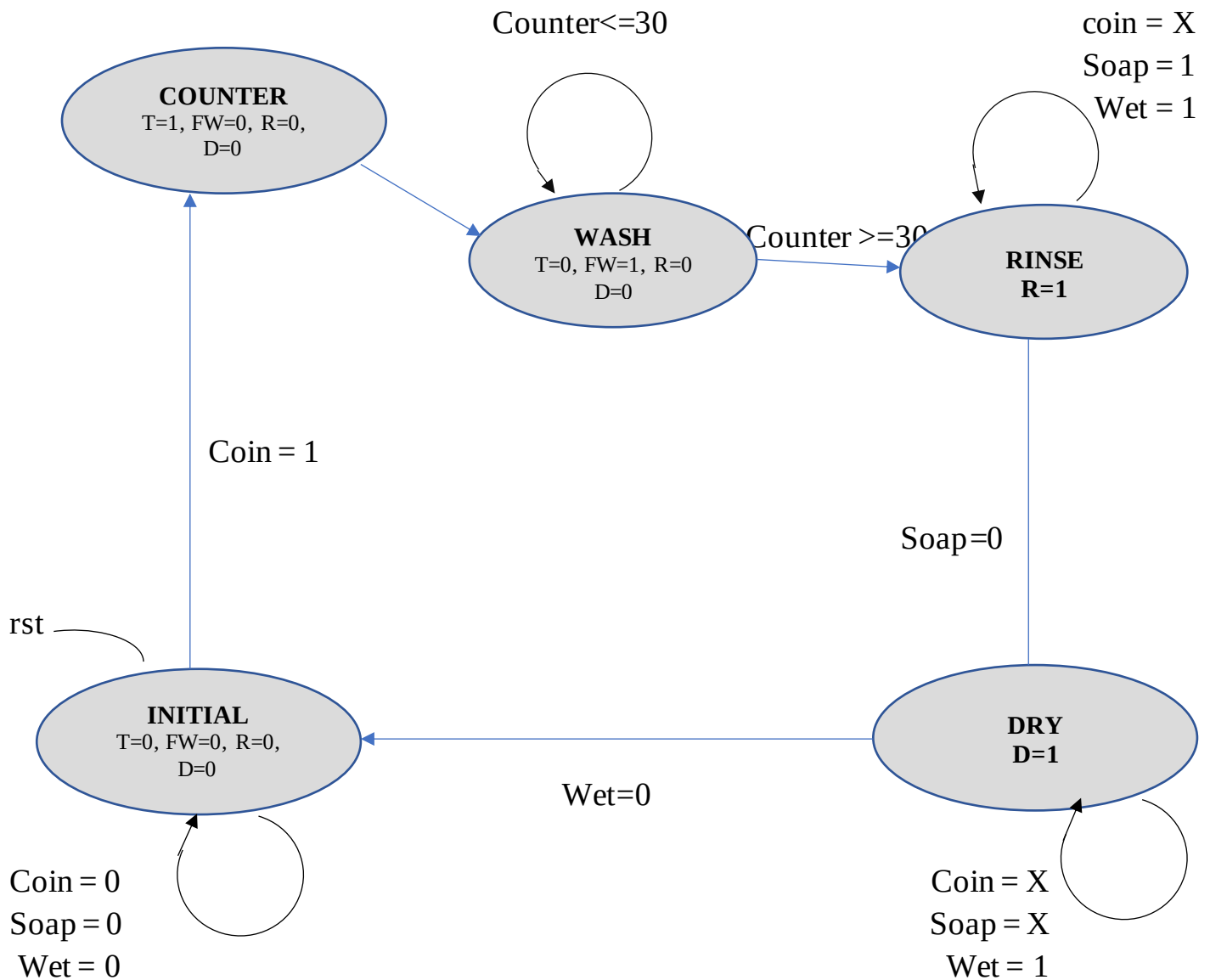
Based on the functional requirements, the entire operation of the washing machine is divided into a set of well-defined states. A **Finite State Machine (FSM)** model is chosen to represent the sequential behavior of the system. Each state corresponds to a specific operation such as idle, washing, rinsing, or drying, and transitions between states occur based on input signals and sensor conditions.

A **state transition diagram** is drawn to visually represent the state flow and transition conditions. From the state diagram, a **state transition table** is derived, which clearly defines the next state and output for every possible input condition. Suitable **state encoding** is then selected to represent each state using binary values.

After defining the state behavior, the design is implemented at the **Register Transfer Level (RTL)** using **Verilog Hardware Description Language (HDL)**. An internal counter is incorporated to control the washing duration accurately. A **testbench** is developed to apply input stimuli and verify the functionality of the design.

Finally, the design is verified using **simulation waveforms** to ensure correct state transitions and output behavior. The verified RTL design is then synthesized to obtain the **synthesis schematic**, confirming the hardware feasibility of the washing machine controller.

STATE TRANSITION DIAGRAM



- The controller initially remains in the **Idle state** waiting for the coin input.
- On receiving the coin, the system moves to the **Counter state** to initiate timing.
- The **Wash state** performs washing for 30 minutes using an internal counter.
- After washing, the system enters the **Rinse state** until soap is removed.
- The **Dry state** continues until the wetness sensor turns off.
- The controller then returns to the **Idle state**.

STATE TRANSITION TABLE AND STATE ENCODING

Present State	INPUT				Next State	OUTPUT			
	Coin	Counter	Soap	Wet		Timer (T)	Fill Water (FW)	Drain-Refill (DR)	Drain (D)
Initial	0	X	X	X	Initial	0	0	0	0
Initial	1	X	X	X	Counter	0	0	0	0
Counter	X	X	X	X	Wash	1	0	0	0
Wash	X	1	X	X	Rinse	0	0	0	0
Wash	X	0	X	X	Wash	0	1	0	0
Rinse	X	X	0	X	Dry	0	0	0	0
Rinse	X	X	1	X	Rinse	0	0	1	0
Dry	X	X	X	0	Initial	0	0	0	0
Dry	X	X	X	1	Dry	0	0	0	1

STATE ENCODING TABLE

State	Q2	Q1	Q0
Initial	0	0	0
Counter	0	0	1
Wash	0	1	0
Rinse	0	1	1
Dry	1	0	0

Present State			Input				Next State				Output		
Q2	Q1	Q0	C	Count	S	W	Q2 ⁺	Q1 ⁺	Q0 ⁺	T	FW	DR	D
0	0	0	0	X	X	X	0	0	0	0	0	0	0
0	0	0	1	X	X	X	0	0	1	0	0	0	0
0	0	1	X	X	X	X	0	1	0	1	0	0	0
0	1	0	X	0	X	X	0	1	0	0	1	0	0
0	1	0	X	1	X	X	0	1	1	0	1	0	0
0	1	1	X	X	0	X	1	0	0	0	0	1	0
0	1	1	X	X	1	X	0	1	1	0	0	1	0
1	0	0	X	X	X	0	0	0	0	0	0	0	1
1	0	0	X	X	X	1	1	0	0	0	0	0	1

Present State			Input				Next State			Output				D inputs					
Q2	Q1	Q0	C	Count	S	W	Q2 ⁺	Q1 ⁺	Q0 ⁺	T	FW	DR	D	D2	D1	D0	Q _n	Q _{n+1}	D
0	0	0	0	X	X	X	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	X	X	X	0	0	1	0	0	0	0	0	0	1	0	1	1
0	0	1	X	X	X	X	0	1	0	1	0	0	0	0	1	0	1	0	0
0	1	0	X	0	X	X	0	1	0	0	1	0	0	0	1	0	1	1	1
0	1	0	X	1	X	X	0	1	1	0	1	0	0	0	1	1			
0	1	1	X	X	0	X	1	0	0	0	0	1	0	1	0	0			
0	1	1	X	X	1	X	0	1	1	0	0	1	0	0	1	1			
1	0	0	X	X	X	0	0	0	0	0	0	0	1	0	0	0			
1	0	0	X	X	X	1	1	0	0	0	0	0	1	1	0	0			

OUTPUT EXPRESSION

Output expression and excitation expressions:

Next State Equations

$$D_2 = f(Q_2, Q_1, Q_0, \text{Inputs}) = Q_2'Q_1'Q_0' C + Q_2'Q_1Q_0' \text{Count} + Q_2'Q_1Q_0 S$$

$$D_1 = f(Q_2, Q_1, Q_0, \text{Inputs}) = Q_2'Q_1'Q_0 + Q_2'Q_1Q_0' + Q_2'Q_1Q_0 S$$

$$D_0 = f(Q_2, Q_1, Q_0, \text{Inputs}) = Q_2'Q_1Q_0 S' + Q_2Q_1'Q_0' W$$

$$T = Q_2'Q_1'Q_0'$$

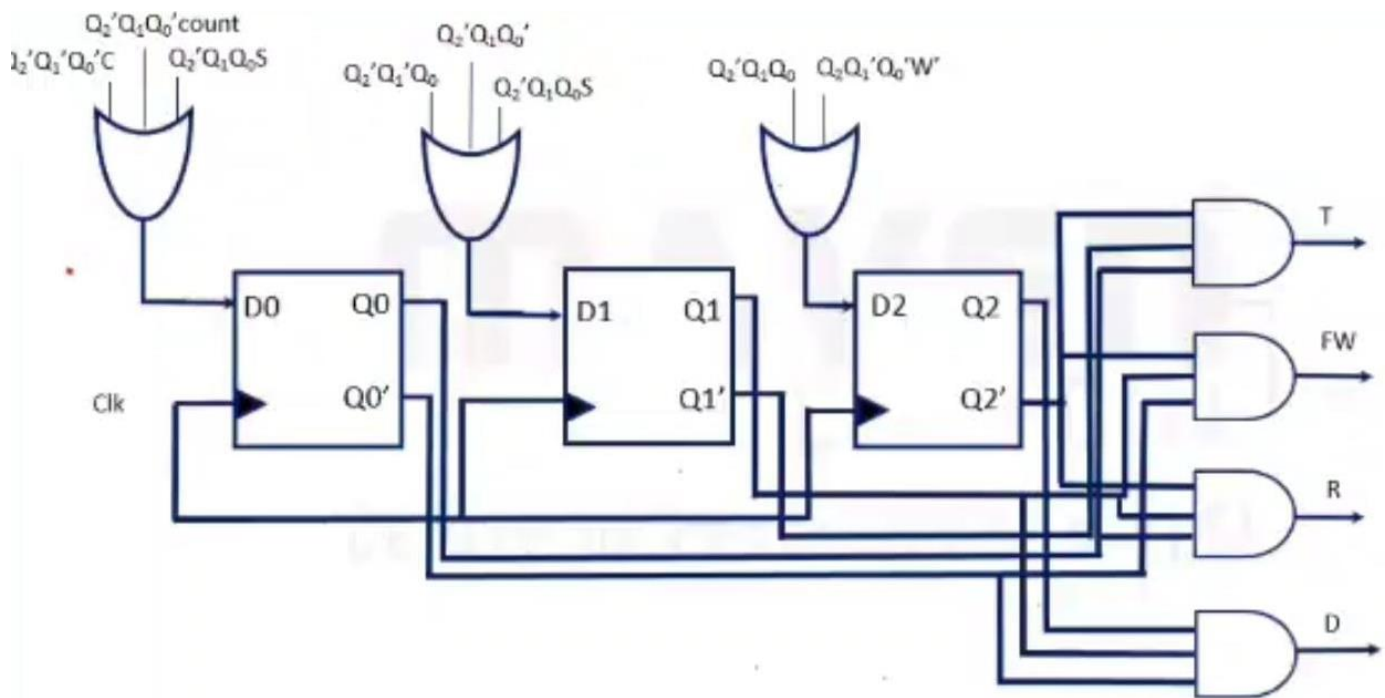
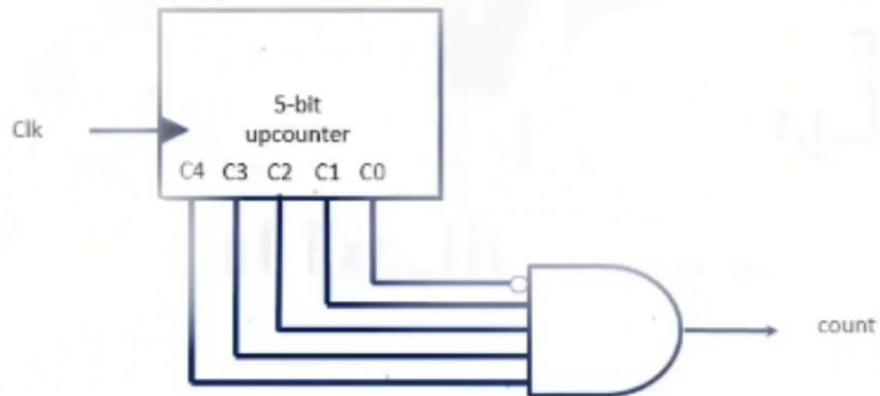
$$FW = Q_2'Q_1Q_0'$$

$$DR = Q_2'Q_1Q_0$$

$$D = Q_2Q_1'Q_0'$$

IMPLEMENTATION

- Counter has to assert the signal 'count' if the count is 30.
- 5 bit timer should be used for this operation along with some basic gates as shown.
- The output of the counter is given as one of the input to the FSM.



RTL DESIGN

RTL CODE: (washer.v)

```
module washer(clk, rst, coin, wet, soap, Timerbegin, Fillwater, Refill, Drain, timer_counter);
```

```
input clk, rst, coin, wet, soap;
```

```
output Timerbegin, Fillwater, Refill, Drain;
```

```
reg [2:0] cst, nst;
```

```
output reg [4:0] timer_counter;
```

```
// state assignment
```

```
parameter IDLE = 3'b000,
```

```
        COUNTER = 3'b001,
```

```
        WASH = 3'b010,
```

```
        RINSE = 3'b011,
```

```
        DRY = 3'b100;
```

```
// Logic for internal counter
```

```
always @(posedge clk)
```

```
begin
```

```
    if (rst)
```

```
        timer_counter <= 0;
```

```
    else if (coin == 1)
```

```
        timer_counter <= timer_counter + 1;
```

```
    else
```

```
        timer_counter <= 0;
```

```
end
```

```
// current state logic
```

```
always @(posedge clk)
```

```
begin
```

```
    if (rst)
```

```
        cst <= IDLE;
```

```
    else
```

```
    cst <= nst;
end

// next state logic
always @(cst or coin or wet or soap)
begin
    case(cst)

        IDLE:
            if (coin == 1)
                nst = COUNTER;
            else
                nst = IDLE;

        COUNTER:
            nst = WASH;

        WASH:
            if (timer_counter > 30)
                nst = RINSE;
            else
                nst = WASH;

        RINSE:
            if (soap == 0)
                nst = DRY;
            else
                nst = RINSE;

        DRY:
            if (wet == 0)
                nst = IDLE;
            else
                nst = DRY;

        default:
```

```
    nst = IDLE;

    endcase
end

// output logic
assign Timerbegin = (cst == COUNTER);
assign Fillwater = (cst == WASH);
assign Refill = (cst == RINSE);
assign Drain = (cst == DRY);

endmodule
```

TESTBENCH

TESTBENCH CODE:(washer_tb.v)

```
module washer_tb();
```

```
reg clk, rst, coin, wet, soap;
```

```
wire Timerbegin, Fillwater, Refill, Drain;
```

```
wire [4:0] timer_counter;
```

```
washer m1(clk, rst, coin, wet, soap, Timerbegin, Fillwater, Refill, Drain, timer_counter);
```

```
// clock generation
```

```
initial
```

```
begin
```

```
    clk = 1'b0;
```

```
    forever #5 clk = ~clk;
```

```
end
```

```
task initialize;
```

```
begin
```

```
    coin = 0;
```

```
    wet = 0;
```

```
    soap = 0;
```

```
end
```

```
endtask
```

```
task reset;
```

```
begin
```

```
    @(negedge clk)
```

```
    rst = 1'b1;
```

```
    @(negedge clk)
```

```
    rst = 1'b0;
```

```
end
```

```
endtask
```

```
task inputs(input c, w, s);
```

```

begin
    @(negedge clk)
    coin = c;
    wet = w;
    soap = s;
end
endtask

task counter;
repeat (31)
begin
    @(posedge clk);
end
endtask

initial begin
    $dumpfile("wave.vcd");
    $dumpvars(0,washer_tb);
end

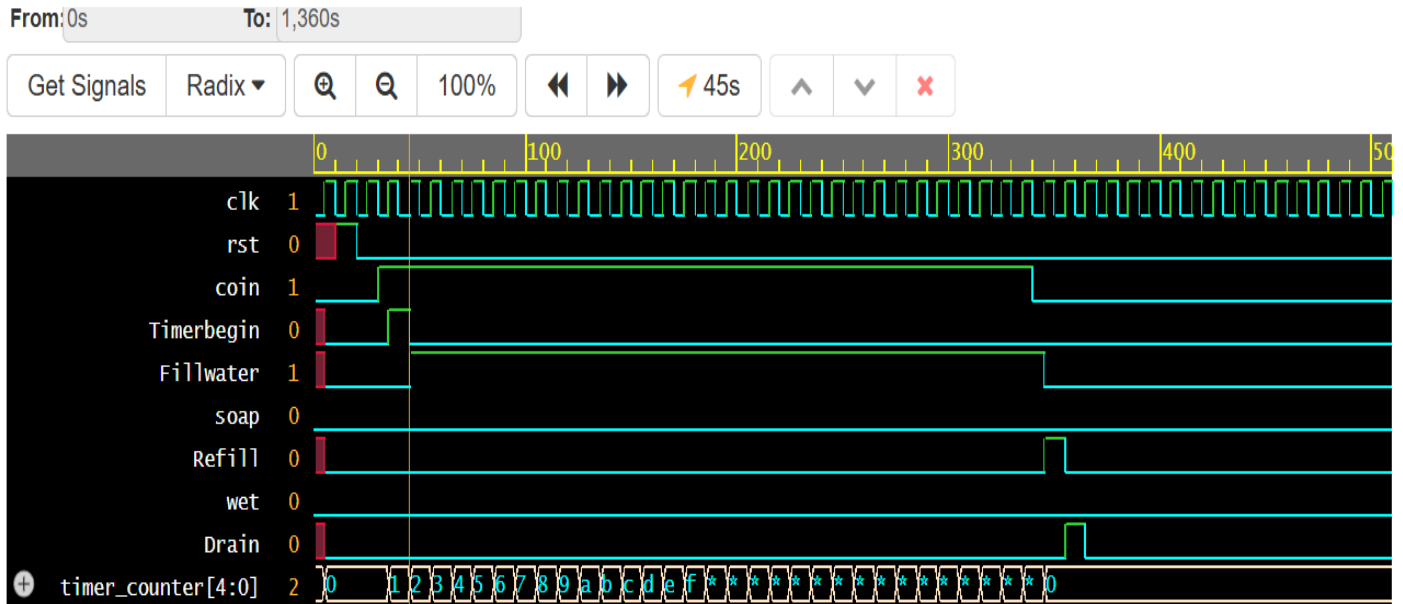
initial
begin
    initialize;
    reset;
    inputs(1,0,0);
    counter;
    inputs(0,0,0);
    inputs(0,0,0);
    inputs(0,0,0);

    #1000 $finish;
end

endmodule

```

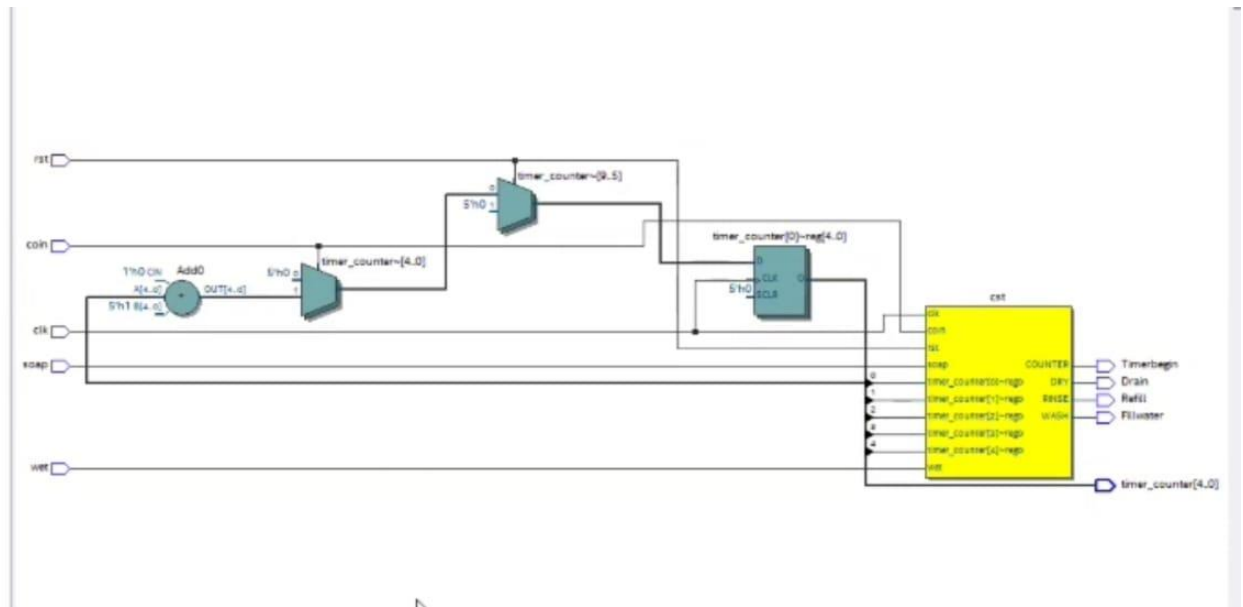
SIMULATION RESULTS



Procedure:

- Open EDA Playground website and create a new simulation workspace.
- Select Verilog HDL as the design language and simulator tool.
- Enter the RTL design code and corresponding testbench code.
- Configure simulation settings and ensure correct top module selection.
- Run the simulation to execute the testbench and generate waveforms.
- Observe signal transitions and verify correct FSM operation.
- Analyze waveform results to confirm expected output behavior.

SYNTHESIS RESULTS



Procedure:

- Open Quartus Prime Lite Edition and create a new project.
- Specify project name, directory, and select the target FPGA device.
- Add the Verilog RTL design file to the project.
- Set the top-level entity corresponding to the washing machine controller.
- Compile the design to perform analysis and synthesis.
- View the synthesized schematic using the RTL Viewer or Technology Map Viewer.
- Verify successful compilation and check for errors or warnings.

CONCLUSION

In this project, a Washing Machine Controller was successfully designed and verified using Finite State Machine (FSM) concepts. The controller effectively automates the washing process by managing operations such as washing, rinsing, and drying based on sensor inputs and timing constraints. The design was implemented at the Register Transfer Level (RTL) using Verilog Hardware Description Language (HDL).

The functional correctness of the controller was verified using a properly designed testbench and simulation waveforms, which confirmed correct state transitions and output behavior. The RTL design was further validated through synthesis, ensuring hardware feasibility and logical correctness. This project provided practical exposure to digital design methodology and FSM-based control systems.

Overall, the project enhanced understanding of IC chip design and verification concepts and demonstrated the real-world application of digital systems in automation, making it a valuable learning experience